Eduard Gorbunov

Tenure-Track Assistant Professor of Statistics and Data Science, MBZUAI, Abu Dhabi, UAE

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At a Glance

METRICS

Google Scholar 4,136 citations
h-index / i10 36 / 56

Since 2021: 3,867 citations, h-index 34, i10-index 52. Metrics checked October 19, 2026.

PUBLICATION RECORD

- 80 listed publications plus 1 book
- 53 conference papers
- 17 journal papers
- 8 arXiv preprints
- 2 reference entries
- 65 talks and posters

COLLABORATION FOOTPRINT

- 113 publication collaborators
- 35 universities/research institutes
- 6 companies/industrial labs

RESEARCH AREAS

- Stochastic optimization
- Distributed and federated learning
- Heavy-tailed noise and clipping
- Variational inequalities
- Adaptive and communication-efficient training

RECOGNITION

- ICML 2024 oral
- NeurIPS 2024 spotlight
- NeurIPS 2020 spotlight
- MOTOR 2026 Best Application Paper Award
- Ilya Segalovich Award, 2019
- NeurIPS 2026 Top Area Chair (Top AC)
- Outstanding reviewer: NeurIPS (2022, 2021, 2020), ICML (2022, 2021), ICLR (2021)

Professional Experience

2025–now	Tenure-Track Assistant Professor, Statistics and Data Science, Mohamed bin Zayed University of Artificial Intelligence.
2024–2025	Research Scientist, Machine Learning Department, MBZUAI.
2022–2024	Postdoctoral Fellow, Machine Learning Department, MBZUAI; hosted by Samuel Horváth and Martin Takáč.
2020–2022	Junior Researcher, MIPT/HSE/Yandex Research; Research Consultant, Mila, Montreal.
2017–2020	Research positions and internships with Peter Richtárik's group, Huawei–MIPT, IITP RAS, RANEPA–MIPT, and KAUST Visual Computing Center.

Education

2021	PhD in Computer Science, Moscow Institute of Physics and Technology. Thesis: *Distributed and Stochastic Optimization Methods with Gradient Compression and Local Steps*. Advisors: Alexander Gasnikov and Peter Richtárik.
2020	MSc in Applied Mathematics and Physics, Moscow Institute of Physics and Technology.
2018	BSc in Applied Mathematics and Physics, Moscow Institute of Physics and Technology.

Research Profile

I develop stochastic and distributed optimization algorithms for machine learning, with emphasis on provable and practical behavior under heavy-tailed noise, communication constraints, local steps, compression, adaptivity, and federated or decentralized systems. Recent work connects clipping, Adam/AdaGrad-type normalization, LMO-based/Muon-type methods, and high-probability and last-iterate analysis to modern training regimes.

Selected Recent Publications

Tags: $*$ equal contribution; † shared senior authorship; mentored/supervised author markers: ^{B} BSc, ^{M} MSc, ^{P} PhD, ^{R} RA/Postdoc, ^{T} Team Member, ^{V} Visiting Student.

1. A. Shestakov^{M}, M. Takáč, E. Gorbunov. **From Optimization to Generalization under Heavy-Tailed Data: The Role of Gradient Clipping**. ICML 2026.
2. E. Shulgin^{V}, M. Awad^{T}, P. Richtárik, E. Gorbunov. **General Analysis of LMO-based Optimizers: Beyond Bounded Variance**. ICML 2026.
3. R. Islamov^{V}, R. Machacek, A. Lucchi, A. Silveti-Falls, E. Gorbunov†, V. Cevher†. **On the Role of Batch Size in Stochastic Conditional Gradient Methods**. ICML 2026.
4. S. Chezhegov^{P}, D. A. Parletta, A. Paudice, E. Gorbunov. **High-Probability Bounds for the Last Iterate of Clipped SGD**. ICLR 2026.
5. S. Khirirat, A. Sadiev^{P}, A. Riabinin, E. Gorbunov, P. Richtárik. **Error Feedback under (L_0, L_1)-Smoothness: Normalization and Momentum**. NeurIPS 2025.
6. S. V. Khah^{M}, S. Chezhegov^{P}, S. Farahmand, S. Horváth, E. Gorbunov. **Differentially Private Clipped-SGD: High-Probability Convergence with Arbitrary Clipping Level**. AISTATS 2026.
7. S. Chezhegov^{P}, Y. Klyukin, A. Semenov, A. Beznosikov, A. Gasnikov, S. Horváth, M. Takáč, E. Gorbunov. **Clipping Improves Adam-Norm and AdaGrad-Norm when the Noise Is Heavy-Tailed**. ICML 2025.

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Mentoring & Service

CURRENT TEAM

- 4 MSc students at MBZUAI
- 1 postdoc at MBZUAI
- 1 current visiting PhD student
- 5 alumni
- Current/recent team members already appear as coauthors in 2026 ICML/UAI papers

EXTERNAL MENTORING

- 27 unique externally mentored publication contributors in the full-CV role markers
- Breakdown: 2 BSc, 14 MSc, 8 PhD, 3 Research Assistant/Postdoc
- Publication outcomes at AISTATS, ICML, ICLR, NeurIPS, CPAL, EMNLP, MOTOR, and IFAC

TEACHING

- AI2100 Multivariable Calculus, BSc Year 2, MBZUAI (Fall 2026; with Roman Bresson and James Brian Pitts)
- ML804 Advanced Topics in Continuous Optimization, PhD, MBZUAI
- MTH702 Optimization, MSc, MBZUAI
- Optimization Methods for Machine Learning, MIPT/MADE
- TA/lecturer contributions to ML712 Distributed and Federated Learning

EDITORIAL SERVICE

- Associate Editor, Statistics and Computing
- TMLR Action Editor
- Area Chair: NeurIPS, ICLR, ICML
- Senior Program Committee Member: AAAI 2027
- Co-organizer: ICOMP
- Reviewer for JMLR, Mathematical Programming, SIMODS, SIOPT, TMLR, NeurIPS, ICML, ICLR, AISTATS, NETYS

Selected Foundational and High-Visibility Publications

Tags: $*$ equal contribution; † shared senior authorship; mentored/supervised author markers: [B] BSc, [M] MSc, [P] PhD, [R] RA/Postdoc, [T] Team Member, [V] Visiting Student.

1. E. Gorbunov*, N. Tupitsa*[R], S. Choudhury[P], A. Aliev[M], P. Richtárik, S. Horváth, M. Takáč. **Methods for Convex (L_0, L_1)-Smooth Optimization: Clipping, Acceleration, and Adaptivity**. ICLR 2025.
2. E. Gorbunov, A. Sadiev[P], M. Danilova, S. Horváth, G. Gidel, P. Dvurechensky, A. Gasnikov, P. Richtárik. **High-Probability Convergence for Composite and Distributed Stochastic Minimization and Variational Inequalities with Heavy-Tailed Noise**. ICML 2024 oral.
3. K. Mishchenko, E. Gorbunov, M. Takáč, P. Richtárik. **Distributed Learning with Compressed Gradient Differences**. Optimization Methods and Software 2025.
4. E. Gorbunov, M. Danilova, A. Gasnikov. **Stochastic Optimization with Heavy-Tailed Noise via Accelerated Gradient Clipping**. NeurIPS 2020.
5. E. Gorbunov, S. Horváth, P. Richtárik, G. Gidel. **Variance Reduction is an Antidote to Byzantine Workers: Better Rates, Weaker Assumptions and Communication Compression as a Cherry on the Top**. ICLR 2023.
6. E. Gorbunov, N. Loizou, G. Gidel. **Extragradient Method: $\mathcal{O}(1/k)$ Last-Iterate Convergence for Monotone Variational Inequalities and Connections with Cocoercivity**. AISTATS 2022.
7. M. Ryabinin*, E. Gorbunov*, V. Plokhotnyuk, G. Pekhimenko. **Moshpit SGD: Communication-Efficient Decentralized Training on Heterogeneous Unreliable Devices**. NeurIPS 2021.
8. E. Gorbunov, K. Burlachenko[P], Z. Li, P. Richtárik. **MARINA: Faster Nonconvex Distributed Learning with Compression**. ICML 2021.
9. E. Gorbunov, D. Kovalev, D. Makarenko[P], P. Richtárik. **Linearly Converging Error Compensated SGD**. NeurIPS 2020 spotlight.
10. E. Gorbunov, F. Hanzely, P. Richtárik. **A Unified Theory of SGD: Variance Reduction, Sampling, Quantization and Coordinate Descent**. AISTATS 2020.

Current Team

- **MSc students at MBZUAI:** Bilal Ashfaq (co-supervised with Nils Lukas), Mohamed Ayman Mohamed Mohamed Awad, Saurabh Singh, Viktor Kovalchuk (co-supervised with Martin Takáč); all 2025/09–present.
- **Postdoc at MBZUAI:** Egor Shulgin (2026/09–present; co-advisor: Eric Moulines).
- **Visiting PhD student:** Taha El Bakkali El Kadi (2026/08–present).
- **Alumni:** Ahmed El Bajdali, Research Engineer (2025/11–2026/09; next: Saptarshi Roy's group, MBZUAI); visitors: Igor Ignashin and Savelii Chezhegov (2026/01–03), Rustem Islamov (2025/12–2026/02), Egor Shulgin (2025/10–11).

Talks, Visibility, and Collaborations

- 65 talks and posters, including an ICML 2024 oral presentation, invited/keynote talks at NETYS, INSAIT, Federated Learning One-World Seminar, MBZUAI Workshop on Collaborative Learning, MLO EPFL, MTL MLOpt, KAUST Rising Stars in AI Symposium, and Russian Optimization Seminar.
- Research visits or collaborations with Mila, KAUST, SIERRA/INRIA, MIPT, EPFL, University of Toronto, HSE, Yandex Research, and other international groups.